

Econ 30065 Microeconomics

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Production and costs

Summary

- Demand side: Demand Curves
- Supply side: Supply Curves?
 - Production Functions
 - Cost Functions
 - Profits Maximization
 - Individual supply curve
- Market Supply and Demand Curve
- Equilibrium: Demand=Supply

Production Technologies

- **Inputs:** the materials, labor, land, or equipment that firms use to produce their outputs
- **Outputs:** the physical products or services a firm produces (Q)
- **Production function:** states mathematically the amount of output a firm can produce from given amounts of inputs using efficient production methods: $Q = F(\text{inputs})$

Production

- Most production processes use many variable inputs: labor, capital, materials, and land
- We assume two homogeneous inputs: labor (L) and capital (K)
- Capital inputs include assets such as physical plant, machinery, and vehicles
- Production function:

$$Q = F(L, K)$$

Production in the Short Run and the Long Run

- **Variable input:** can be adjusted over the time period being considered
- **Fixed input:** cannot be adjusted over the time period being considered
- **Short run:** a period of time over which one or more inputs is fixed
- **Long run:** a period of time over which all inputs are variable
 - Length of long run depends on the production process being considered:
Auto manufacturer may need years to build a new production facility but software firm may need only a month or two to rent and move into a new space

Production

We look at short run: K fixed $\rightarrow Q=F(L)$

➤ $Q=AL^\alpha$

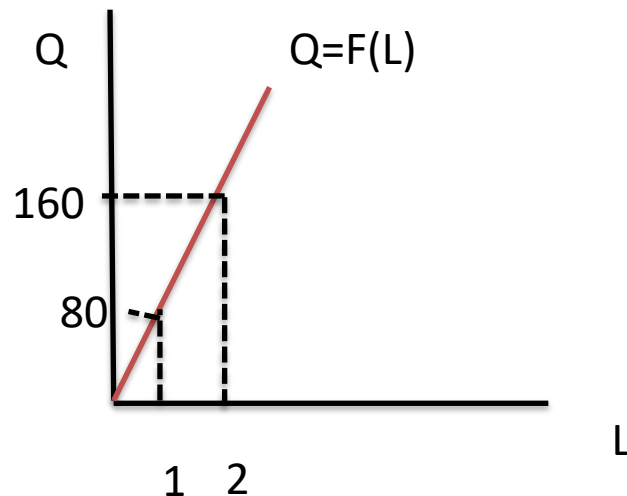
➤ $A>0$ technology level

➤ $\alpha > 0$ L productivity measure

➤ ($\alpha=1$ linear production function, $\alpha < 1$ concave, $\alpha > 1$ convex)

Production Function with 1 variable input: an example

- Example: $Q=F(L)=80L$
 - Q is quantity of output, L is quantity of labor
 - Substitute different amounts of L to see how output changes as the firm hires more workers
 - If $L=1 \Rightarrow Q=80$;
 - If $L=2 \Rightarrow Q=160$;



Average product with one input

➤ $AP_L = F(L)/L$

➤ $Q = AL^\alpha$

➤ $AP_L = A L^{\alpha-1}$

➤ $\alpha = 1 \rightarrow AP_L = A$ constant

➤ $\alpha < 1 \rightarrow AP_L = A L^{\alpha-1}$ decreasing

➤ $\alpha > 1 \rightarrow AP_L = A L^{\alpha-1}$ increasing

➤ Example $Q = 80L$

➤ $AP_L = 80L/L = 80$

Marginal product with one input

- MP captures the additional output we can get for each additional unit of input when we increase the input by the smallest possible amount.

$$MP_L = \frac{\Delta Q}{\Delta L} = \frac{F(L, K) - F(L - \Delta L, K)}{\Delta L}$$

or

$$MP_L = \partial Q / \partial L$$

Marginal product with one input

$$Q = AL^\alpha$$

➤ $MP_L = A \alpha L^{\alpha-1}$

➤ $\alpha = 1 \rightarrow MP_L = A$ constant

➤ $\alpha < 1 \rightarrow MP_L = A \alpha L^{\alpha-1}$ decreasing

➤ $\alpha > 1 \rightarrow MP_L = A \alpha L^{\alpha-1}$ increasing

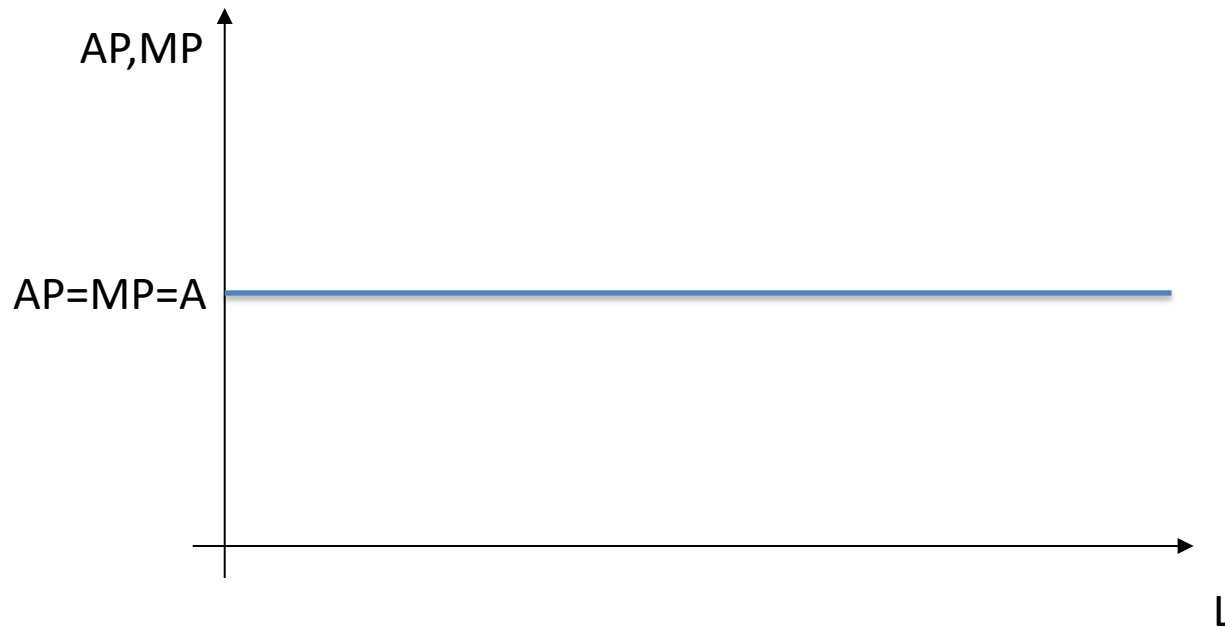
➤ Example $Q = 80L^{0.5}$

➤ $MP_L = dQ/dL = 40/\sqrt{L}$

Average product and marginal product

$$Q=AL^\alpha$$

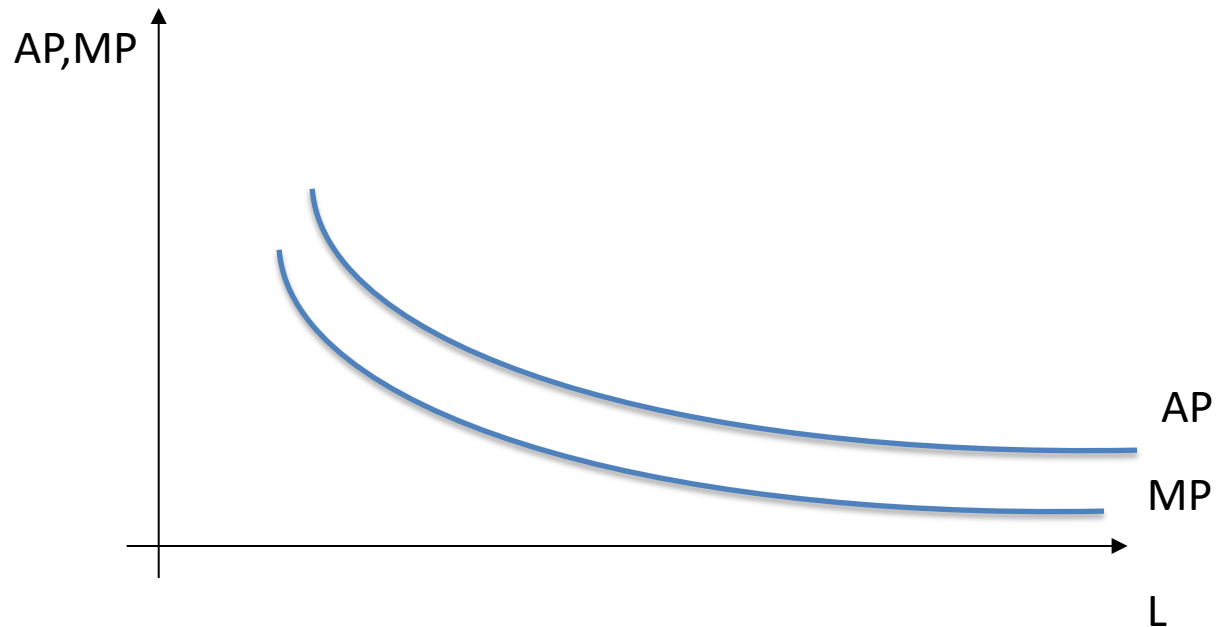
➤ $\alpha=1 \rightarrow AP_L=MP_L$ constant



Average product and marginal product

$$Q = AL^\alpha$$

$$\triangleright \alpha < 1 \rightarrow AP_L > MP_L$$



Total Cost

- Firm's total cost is the expenditure required to produce a given level of output in the most economical way
- Taking the amount that the firm wants to produce as given (this choice will be the subject of our next topic 'profit maximization'), what's the most economical way the firm can produce this amount?

How to find cost functions?

➤ How much does it cost to produce Q ?

➤ Cost function: $TC(Q) = wL(Q) + rK(Q)$,

where w cost of labor, r cost of capital, $L(Q)$ and $K(Q)$ are how much K and L are necessary to produce Q

➤ Short run cost function: K is fixed, FC given, 1 variable input

➤ Find how much L is needed to produce Q (inverse production function)

➤ $TC(Q) = wL(Q) + FC$

➤ Where $FC = rK_{\text{fixed}}$

How to find cost functions?

➤ $Q = AL^\alpha$

➤ $L = (Q/A)^{1/\alpha}$

➤ $TC(Q) = w\left(\frac{Q}{A}\right)^{1/\alpha} + FC$

➤ Example:

➤ $Q = 80L^{0.5}$, $w = 64$, $FC = 100$

➤ $L = (Q/80)^2$

➤ $TC(Q) = 64\left(\frac{Q}{80}\right)^2 + 100$

Types of Costs

- **Short run total cost (TC):**
 - **Variable cost (VC):** costs of inputs that vary with the firm's output level
 - **Fixed cost (FC):** costs of inputs whose use does not vary with the firm's output level and cannot be varied

Types of Costs

- Variable cost function:

$$\text{Variable Cost} = VC(\text{Output})$$

- Fixed cost function:

$$\text{Fixed Cost} = FC$$

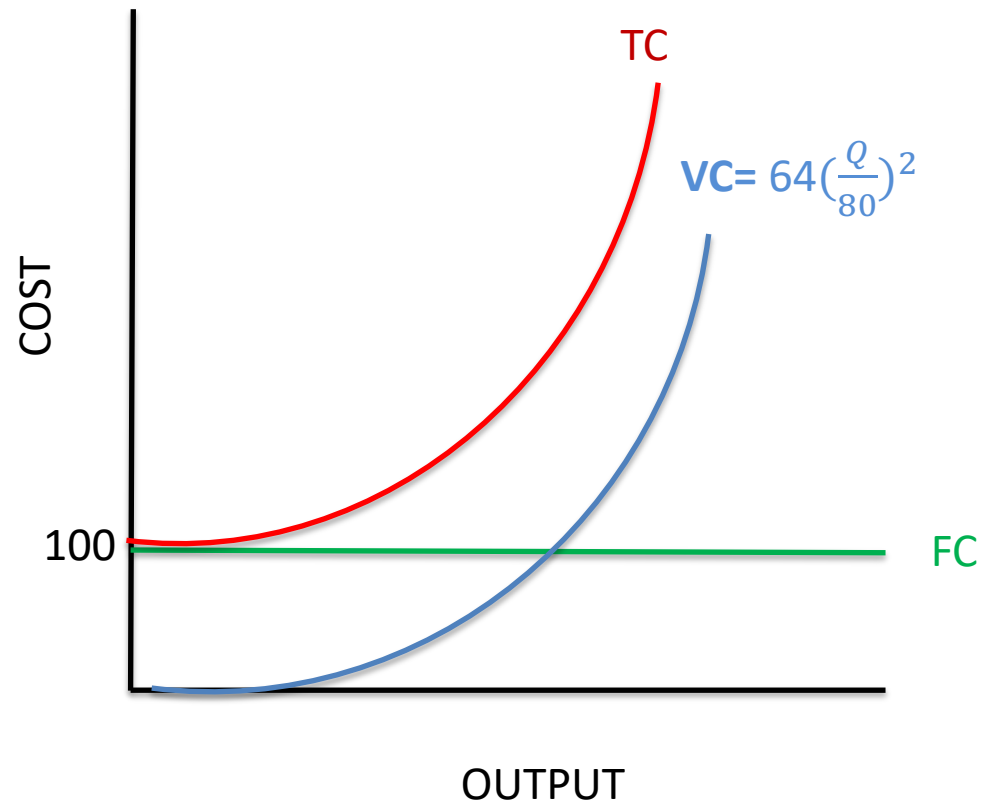
- Total cost function (includes fixed costs):

$$\text{Total Cost} = FC + VC(\text{Output})$$

Fixed, Variable, and Total Cost

Cost curve is equal to the variable cost plus the fixed cost curves:

$$TC(Q) = 64\left(\frac{Q}{80}\right)^2 + 100$$



Average and Marginal cost

- **Average cost:** cost per unit of output produced, averaged over all units produced

$$AC = \frac{C(Q)}{Q}$$

- **Marginal cost:** extra cost the firm incurs per unit of output added

$$MC = \frac{\Delta C}{\Delta Q} = \frac{C(Q) - C(Q - \Delta Q)}{\Delta Q}$$

or

$$MC = \frac{\partial C(Q)}{\partial Q}$$

Average Cost

- Average Variable Cost

$$AVC = \frac{VC(Q)}{Q}$$

- Average Fixed Cost

$$AFC = \frac{FC(Q)}{Q}$$

Average cost:
 $AC = AVC + AFC$

Average variable cost

$$AVC(Q) = \frac{W}{A^{1/\alpha}} Q^{\frac{1}{\alpha}-1}$$

Example:

$$TC(Q) = 64\left(\frac{Q}{80}\right)^2 + 100$$

$$AC(Q) = 64\left(\frac{1}{80}\right)^2 Q + 100/Q$$

Average and Marginal cost

- **Average cost:** cost per unit of output produced, averaged over all units produced

$$AC = \frac{C(Q)}{Q}$$

- **Marginal cost:** extra cost the firm incurs per unit of output added

$$MC = \frac{\Delta C}{\Delta Q} = \frac{C(Q) - C(Q - \Delta Q)}{\Delta Q}$$

or

$$MC = \frac{\partial C(Q)}{\partial Q}$$

$$MC(Q) = W/MP_L$$

Marginal cost

$$MC(Q) = \frac{1}{\alpha} \frac{W}{A^{1/\alpha}} Q^{\frac{1}{\alpha}-1}$$

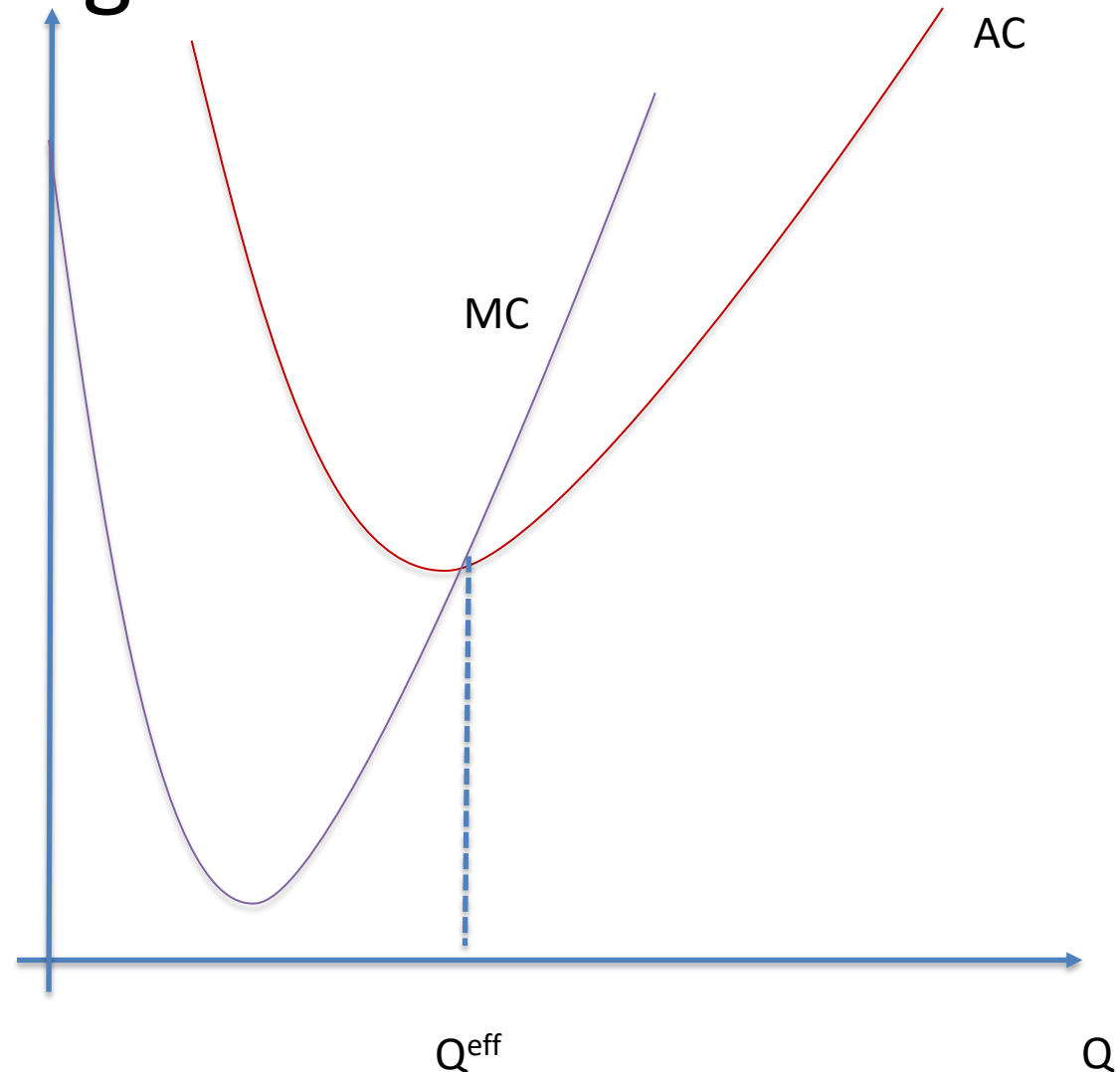
Example:

$$TC(Q) = 64 \left(\frac{Q}{80} \right)^2 + 100$$

$$MC(Q) = 128 \left(\frac{1}{80} \right)^2 Q$$

Relationship between Average and Marginal Cost

- AC curve is upward sloping at Q if $MC > AC$, downward sloping if $MC < AC$, and neither rising nor falling if $MC = AC$
- MC always crosses the AC from below at the efficient scale of production: i.e at min AC



costs

